Reflective Journal – Module 7: Model Evaluation

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1. Introduction

In this lab, I learned how to evaluate a machine learning model using more than just accuracy. I continued working with the Titanic dataset and used Logistic Regression to predict whether passengers would survive. The lab helped me understand how the confusion matrix, precision, recall, F1-score, and cross-validation provide a clearer picture of a model's performance.

2. Description of the Experience

I loaded the Titanic dataset using Pandas, converted the Sex column into numerical values, and filled missing Age values with the median. I used Age, Pclass, Sex, and Fare as features to train the Logistic Regression model. After making predictions, I created a confusion matrix heatmap and generated a classification report. My report showed 80% accuracy. For the Died class, precision was 0.82 and recall was 0.86. For the Survived class, precision was 0.78 and recall was 0.73. I also completed five-fold cross-validation, which produced an average score of 78.68% and a standard deviation of 0.0094.

3. Personal Reflection and Learning

Before this lab, I mainly used accuracy to judge how well a model performed. I learned that accuracy does not show every type of mistake a model can make. The confusion matrix helped me compare the model’s predictions with the actual results. Precision measures how often the model’s positive predictions are correct, while recall measures how many actual positive cases the model identifies. The F1-score combines precision and recall into one measure. I also learned that the best metric depends on the situation. Precision is important when false alarms need to be avoided, while recall is more important when missing a real case could have serious consequences. Finally, cross-validation helped me understand why testing a model on different parts of the dataset provides a more reliable picture of its performance.

4. Challenges, Improvements, and Questions

One challenge was understanding the difference between precision and recall and deciding when each metric is more important. Connecting them to examples such as spam detection and medical screening made the difference easier to understand. I also learned to look at the average cross-

validation score and its standard deviation rather than relying on a single test result. In the future, I would like to compare different models, try different features, and see whether their evaluation scores improve. I also want to learn more about how false positives and false negatives affect decisions in real-world applications.

5. Conclusion

Overall, this lab taught me that evaluating a machine learning model requires looking at several metrics and understanding what each one means. My biggest takeaway is that a model can have good accuracy but still make important mistakes, so it is necessary to examine precision, recall, the F1-score, and cross-validation results. I feel more confident interpreting evaluation results and understand that reliable machine learning involves testing carefully, considering the cost of mistakes, and finding ways to improve the model.